

NutriAI

Automated Personalised Diet Plan Builder — Technical Brief

BAX-423 Big Data · Spring 2026 · Final Project

1. System Overview

NutriAI generates clinically-aware 7-day, 3-meal-per-day nutrition plans from 10,190 USDA FoodData Central records. Given a user profile — clinical conditions, allergens, dietary preferences, and calorie targets — the system filters the food pool, applies semantic diversity ranking via FAISS embeddings, and produces a complete plan with macro/micronutrient analysis against NIH RDA in under 60 seconds.

Layer	Component	Technology
Data Ingestion	USDA FoodData Central API → SQLite	Python requests, sqlite3
Allergen Index	Probabilistic allergen pre-screening	Bloom Filter (pybloom-live)
Semantic Index	Food embedding + similarity search	FAISS + sentence-transformers
Filter Engine	Clinical, allergen, dietary constraints	pandas, regex
Meal Planner	7-day × 3-meal generation + diversity	Custom greedy + FAISS ranking + Stochastic Shuffling
Nutrient Analyser	Macro/micro totals vs NIH RDA	pandas, numpy
User Interface	Profile builder, plan display, analytics	Streamlit + Plotly
Hosting	Live public deployment	Streamlit Community Cloud

2. Data Pipeline

Food data is ingested from four USDA data types: SR Legacy, Survey (FNDDS), Foundation, and Branded, using format=full to include the ingredients field required for accurate allergen tagging. The pipeline extracts 14 macro/micronutrient fields and applies whole-word regex tagging for 9 allergen classes, FODMAP triggers, GERD triggers, and 4 dietary categories. Nutrient extraction handles two USDA response formats: SR Legacy and Foundation return values under 'amount'; Branded and FNDDS return values under 'value' — both keys are tried to ensure full micronutrient coverage.

3. BAX-423 Techniques

Technique 1 — FAISS Semantic Food Embeddings: All 10,190 foods are encoded into 384-dimensional vectors using all-MiniLM-L6-v2, combining name, category, and ingredient text. L2-normalised vectors are indexed with FAISS IndexFlatIP (cosine similarity via inner product), built once and loaded from disk in under 1 second. In the planner, candidates are ranked by dissimilarity to the previous anchor food (invert=True), chaining diversity across all 21 meals. diversity_score() computes mean pairwise dissimilarity. *Scoring Fallback Safeguard:* If underlying database category strings are uniform or unpopulated in a subset, the engine dynamically branches to unique food product names to ensure accurate, non-zero variety tracking.

Technique 2 — Bloom Filter Allergen Index: A Bloom filter (error_rate=0.001) is built per allergen class over unsafe fdci_ids at startup. No false negatives: unsafe foods are always caught. False positives are confirmed by the DB column before exclusion.

Method	Time (ms) / 10,000 lookups	vs Bloom
Bloom Filter	338.5	1.0×
Naive DataFrame query	7,875.3	23.3×
Agreement rate	100.0000%	—

Table 1 — Bloom filter vs. naive DataFrame allergen lookup (10,000 lookups)

4. Clinical Filter Engine

The FilterEngine applies 9 sequential passes to the food pool for a given UserProfile. Each exclusion is logged with a human-readable reason, powering the Explain tab in the UI.

Pass	Condition	Logic
1	Diet type	Skip for non_veg; whitelist is_vegan / is_vegetarian / is_pescatarian
2	Allergens	Bloom pre-screen + DB column confirmation; 9 allergen classes
2b	Name safety net	Whole-word name sweep for allergens where ingredients string is empty
3	IBS / FODMAP	Exclude is_high_fodmap=1 + broad runtime regex sweep for aromatics and triggers (garlic, onion, milk)
4	GERD	Exclude is_gerd_trigger=1 + explicit name sweeps for high pastry fats, chocolate, mint, and fried items
5	Diabetes	Name keyword sweep: pie, cake, pastry, candy, soda, juice, waffle, etc. (USDA has no GI data)
6	Hypertension/DASH	aggressive 1.5 mg/kcal sodium-to-calorie density filtration pass + high-sodium processed keyword sweeps
7	Cultural	Pork, beef, alcohol keyword exclusions with explicit ambiguous 'meat' safeguards
8	Data quality	Drop foods with no calorie data

5. Meal Planner & Diversity Engine

The MealPlanner generates plans in three stages: (1) pool partitioning by meal slot, (2) FAISS-guided diverse food selection with macro-balance scoring, and (3) a calorie top-up loop. *Scoping Guard*: The planner completely isolates intra-day tracking arrays from global weekly collections via read-only clones, eliminating variable mutation leaks and preventing pool starvation. To break deterministic sorting traps, a *Stochastic Shuffle Injector* samples randomly from the top 6 optimal candidates. *Emergency Relief Valve*: Under heavily stacked profiles, if an exclusive nutritional pool starves midway, the engine automatically drops multi-day historical bans while maintaining active same-day allergen/dietary boundaries, guaranteeing a full 21/21 meal delivery.

6. Persona Test Results

Four clinical personas validate the system end-to-end across distinct condition, allergen, and diet combinations.

Persona	Conditions	Diet	Allergens	Safe Foods	Diversity	Time	Pass?
Priya	IBS-D	Vegetarian	Dairy	4,461	0.95	2.8s	PASS
Ravi	GERD, Celiac	Non-veg	Gluten	7,150	0.96	2.2s	PASS
Mei	Type 2 Diabetes	Vegan	Tree nuts	2,800	0.94	2.4s	PASS
James	Hypertension	Pescatarian	Soy	3,855	0.95	2.3s	PASS

Table 2 — Persona validation results

7. Nutrient Analysis

The NutrientAnalyser computes daily macro/micronutrient totals against NIH DRI. Days below 80% RDA are flagged; sodium is flagged above the target cap. Macro profiles scale dynamically in the DB interface layer (e.g., fats compressed to 22% for GERD, carbs to 35% for Diabetes), forcing the planner scoring engine to avoid adverse items. Nutrient calculations handle two USDA field schemas ('amount' and 'value') seamlessly to ensure accurate tracking.

7b. User Interface & Deployment

The Streamlit UI provides three tabs. The **7-Day Plan** tab shows a core metrics header followed by a full-width day grid with breakfast, lunch, and dinner columns. Meal cards show cleaned, reformatted food names, calorie counts, and a P/C/F gram breakdown. The **Nutrition Analysis** tab maps a Plotly bar chart of 13 tracked nutrients as % of RDA and displays interactive daily trend lines. The **Explain** tab queries the ExclusionLog live, providing instant transparency into filtered items.

8. Limitations & Future Work

Limitation	Impact	Mitigation / Future Work
No USDA glycaemic index data	Medium	Diabetes filter uses robust name-based keyword proxy arrays (pie, pastry, candy, syrup)
Missing micronutrient data	Low	NaN handled gracefully; multi-key search unified 'value' vs 'amount' field schemas across types
Raw ingredient contamination	Medium	RAW_INGREDIENT_RE (45+ strict patterns) applied; minor commodity leakages handled in meal top-up passes
Meal composition quality	Medium	Stochastic Candidate Shuffling breaks optimization traps; food category balancing is next iteration
Sodium over-cap bounds	Medium	Hyper-aggressive sodium-to-calorie density caps applied; relief valve triggers on extreme pool starvation
No portion size specification	Medium	Nutrients computed per 100g; future work: integrate serving size mapping records

9. References

1. USDA FoodData Central API. <https://fdc.nal.usda.gov/api-guide.html>

2. NIH Dietary Reference Intakes. <https://www.ncbi.nlm.nih.gov/books/NBK56068>

3. Monash University FODMAP Diet. <https://www.monashfodmap.com>

4. Johnson et al. (2019). Billion-scale similarity search with GPUs. IEEE Trans. Big Data.

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